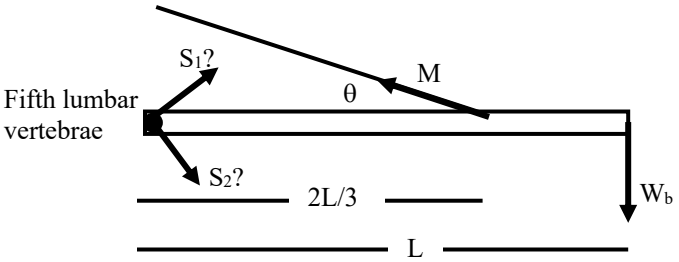
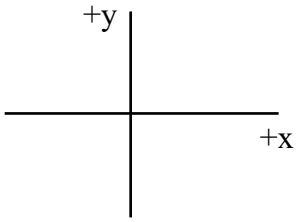
Group Problem 5 Solution



Know:
W_b
L
$\theta = 2^o$



What is the force on the bar at one end (S) as a function of the weight of the box (W_b) on the other end for a known length (L), and angle of the muscle supporting the back (θ)?

Just as the person picks up the box, assume the spine is not accelerating. That means the sum of the forces on the bar is zero and the sum of the torques on the bar also is zero.

Assume the bar is has negligible weight.

The muscle must pull on the spine to lift the back because muscles can only pull. That means the muscle must be above the spine to pull it up.

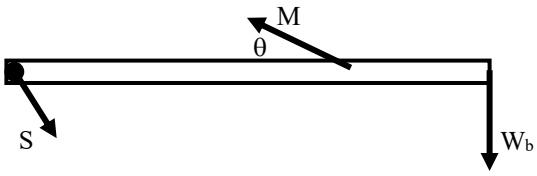
Need to determine the direction of S. It must have a component in the +x direction to balance the -x component of M using the designated coordinate system. That means there are two possibilities for S.

To determine which of the 2 possibilities is correct for S, find the one that gives 0 torque around all pivot points.
Choose the point where W_b acts as the pivot point. M would cause the bar to rotate clockwise. To get 0 torque, S must cause the bar to rotate counterclockwise. That would mean S_2 is in the correct direction.
Check by taking the pivot point where M attaches to the bar. W_b would cause the bar to rotate clockwise so to get 0 torque, S must cause the bar to rotate counterclockwise. This again requires S_2, so the correct direction of S is S_2.

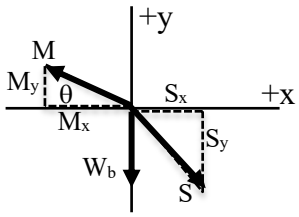
Approach

Assume the bar is in equilibrium for a constant rotation.
The sum of the forces in any direction is zero.
The sum of the torques around any pivot point is zero.
Use a horizontal and vertical coordinate system.
Choose the fifth lumbar vertebrae position as the pivot point because that force (S) is not known.
Choose clockwise rotation as positive.

Free Body Diagram of Bar



Force Diagram of Bar



Add torques: $\sum \tau = -M_y \frac{2L}{3} + W_b L = 0$

Add x force components: $\sum F_x = S_x - M_x = 0$

Add y force components: $\sum F_y = -S_y + M_y - W_b = 0$

Force components: $\sin\theta = \frac{M_y}{M}$ and $\cos\theta = \frac{M_x}{M}$

torques: $-M \sin\theta \frac{2L}{3} + W_b L = 0$

x forces: $S_x - M\cos\theta = 0$

y forces: $-S_y + M\sin\theta - W_b = 0$

S components: $S_x^2 + S_y^2 = S^2$

Target: S

Plan: unknowns

Need S S

$S_x^2 + S_y^2 = S^2$ [1] S_x, S_y

Need S_x

$S_x - M\cos\theta = 0$ [2] M

Need S_y

$-S_y + M\sin\theta - W_b = 0$ [3]

Need M

$-M\sin\theta\frac{2L}{3} + W_b L = 0$ [4]

4 unknowns and 4 equations,
OK to solve.

Execute the Plan:

Solve [4] for M

$$-M\sin\theta\frac{2L}{3} + W_b L = 0$$

$$-M\sin\theta\frac{2}{3} + W_b = 0$$

$$-\frac{2}{3}M\sin\theta = -W_b$$

$$M = \frac{W_b}{\frac{2}{3}\sin\theta}$$ put into [3 and 2]

[2] $$S_x - \frac{W_b}{\frac{2}{3}\sin\theta}\cos\theta = 0$$

Put M into [3]. Solve for S_y

$$-S_y + \frac{W_b}{\frac{2}{3}\sin\theta}\sin\theta - W_b = 0$$

$$-S_y = -\frac{3}{2}W_b + W_b$$

$$S_y = \frac{1}{2}W_b$$ Put into [1]

[1] $$S_x^2 + \left(\frac{1}{2}W_b\right)^2 = S^2$$

Solve [2] for S_x

Put into [1]

$$S_x = \frac{W_b}{\frac{2}{3}\sin\theta}\cos\theta$$

$$\left(\frac{W_b}{\frac{2}{3}\sin\theta}\cos\theta\right)^2 + \left(\frac{1}{2}W_b\right)^2 = S^2$$

$$\left(\left(\frac{3}{\tan\theta}\right)^2 + 1\right)\left(\frac{1}{2}W_b\right)^2 = S^2$$

$$\boxed{\frac{1}{2}W_b\sqrt{\left(\left(\frac{3}{\tan\theta}\right)^2 + 1\right)} = S}$$

Evaluate:
Check Units:
There are only numbers with no units in the square root. W_b is a force and S is a force, so the units check.

If the weight being lifted, W_b, increases, the force on the spine increases. This is reasonable.

If θ were 0, S would be infinite. This is reasonable since θ = 0 means M is along the rod and there is no value of S that would keep the rod in equilibrium.

Put in $\theta = 2^o$

$$\frac{1}{2}W_b\sqrt{\left(\left(\frac{3}{\tan 2°}\right)^2 + 1\right)} = S$$

$$\frac{1}{2}W_b\sqrt{\left(\left(\frac{3}{0.035}\right)^2 + 1\right)} = S$$

$$43\,W_b = S$$

The units check since weight is a force and S is a force.
The force on the fifth lumbar vertebrae is very much larger than the weight of the box. However, since I know this type of lift will injure the back, this is not unreasonable. A force smaller than the box weight or a thousand times larger would be unreasonable.